Research Paper on the Fundamentals of 3D Perspective Projection In Computer Graphics

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October 10, 2025

Abstract

Add abstract

Contents

1 Introduction **2**

2 Ray Tracing **3**
2.1 Coordinates and Vectors 3
2.2 Viewports and Rays 7
2.3 A Ray Tracer Implementation 9

1 Introduction

When you look at the world through your eyes, you are seeing a 3-dimensional space. In contrast, the image that you see possesses only 2 dimensions, and yet, we recognize our view as a 3D one. Through this view, one can still perceive the depth of such an image, retrieving some of the information lost in the process of flattening the view into 2 dimensions. This phenomenon reveals the possibility of projecting a 3D space onto a 2D plane, allowing us to "emulate" the existence of a 3D space behind a 2D plane (much like looking through a window to look at the outside world; a detailed enough 2D image on the window would be indistinguishable from the outside view).

In truth, this possibility is taken advantage of to form the foundations of modern computer graphics. When it comes to video games, computer generated imagery, or 3D animation, computers must simulate all the physical processes of the real world that produce the light and resulting image that reaches your eyes. Although the entirety of the physical world cannot possibly be simulated within the constraints of modern computing capabilities, mathematical and physical abstractions can be made in order to simplify this process and produce near true to life results. These abstractions form the basis of 3D Computer Graphics.

In modern computing, knowing the fundamental concepts and processes behind projecting a 3D image has been made unnecessary to know. With the release of many software solutions and libraries that aid in the development of appropriate visuals, the modern day developer does not need to grasp the underlying mechanics behind 3D Computer Graphics in order to produce satisfactory results.

However, the benefits of learning the fundamental process behind this widely used field are essential for those who wish to become true experts not tied down by any specific engine or library. Factors such as performance, scalability, and customizability are reasons that justify the process of learning the fundamentals. Above all else, those who understand are better equipped to take advantage of and develop graphical toolkits. As such, it is pertinent to understand the foundations that make up 3D Computer Graphics for those who wish to become true masters of the field.

Add real-world examples of mentioned advantages

This research paper sets out to document the process of learning the fundamentals of 3D Computer Graphics, with real code examples mapping out the learning process. As such, it aims to serve as a useful guide that directs readers to explore the bare-bones fundamentals of 3D Computer Graphics.

Add limitations of this paper

2 Ray Tracing

2.1 Coordinates and Vectors

In order to represent a 3-dimensional space, we must first develop the tools used for defining a 3D space.

A **cartesian coordinate system** is a system of numerically representing space with a set of coordinates. n-dimension spaces require n sets of coordinates to represent. As such, a 3-dimension cartesian coordinate system requires 3 sets of coordinates, commonly referred to as x, y, and z.

A **point** is a single position in space, represented by a set of coordinates, and notated by $(C_1, C_2, C_3, \ldots, C_n)$. A 3D cartesian coordinate system places the center, or the origin point at $(0, 0, 0)$. Each dimension, referred to as an axis, has a direction: traveling towards the direction of an axis is known as traveling positively, while traveling against the direction of an axis is known as traveling negatively, with the corresponding coordinates increasing or decreasing appropriately.[1]

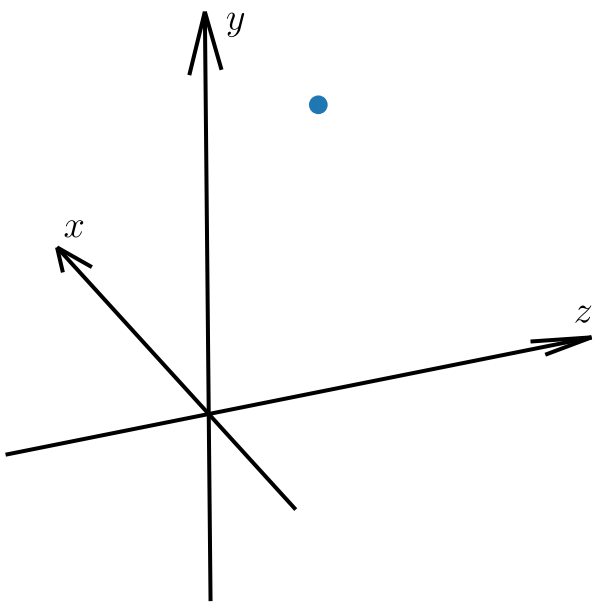


Figure 1: The point (1, 1, 1) in a 3D cartesian coordinate system

The directions of each axis in relation to each other can be remembered using the right hand rule.

Axis configurations corresponding to the left hand also exist, however the right-handed system is more common.

[1] Eldridge, *Cartesian coordinates*.

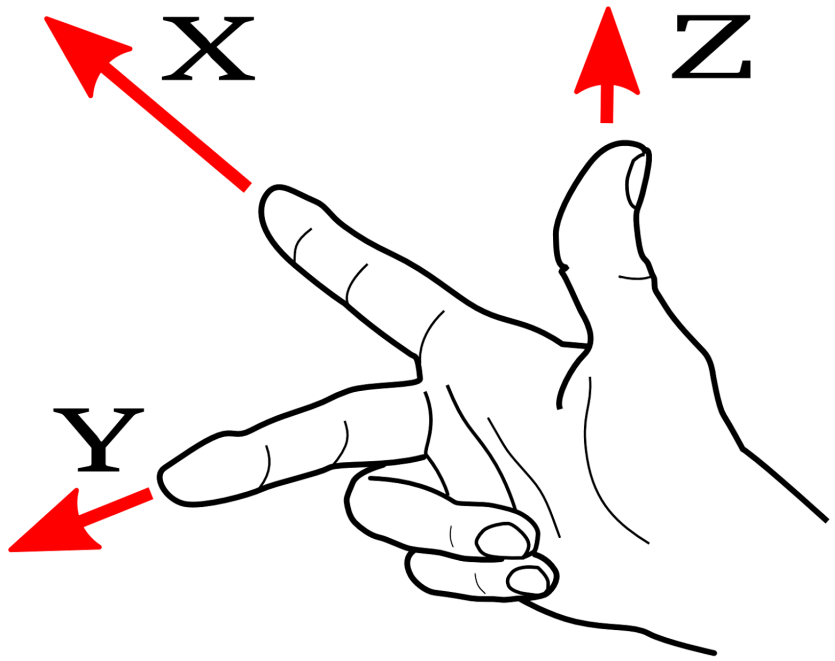


Figure 2: Right-hand rule

Vectors are the differences between points. Vectors can be used to represent many things, such as motion and direction. At its simplest, it can be intuitively thought of as a set of components that represent a certain magnitude of motion in a certain direction. It can also be thought of as the concept of a slope used in 2D cartesian coordinate systems expanded for n-dimensions. Visually, it is a line connecting two points in space (Figure 3).

n-dimension vectors can be notated as a set of n components $(C_1, C_2, C_3, \ldots, C_n)$. Much like the slope of a line in 2 dimensions, vectors can be thought of as the ratio between each of the components' deltas (differences in magnitude). For example, take the 3D vector $(1, 2, 3)$: when x is increased by 1, y is increased by 2, and z is increased by 3 (Figure 4).[2]

[2]Encyclopedia Britannica, *vector*.

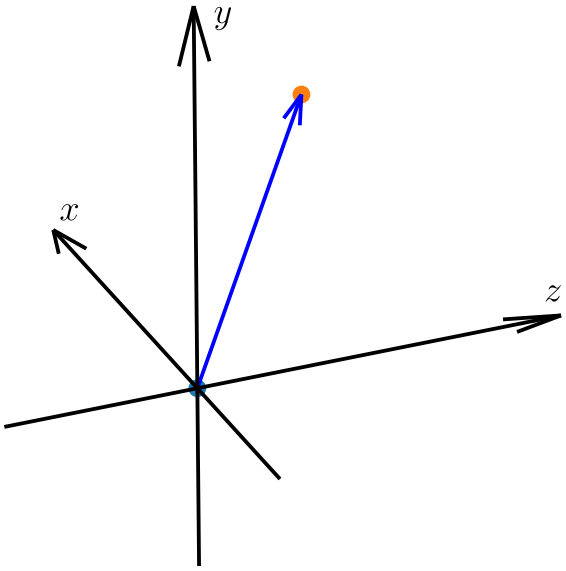


Figure 3: A vector from points $(0, 0, 0)$ to $(1, 1, 1)$

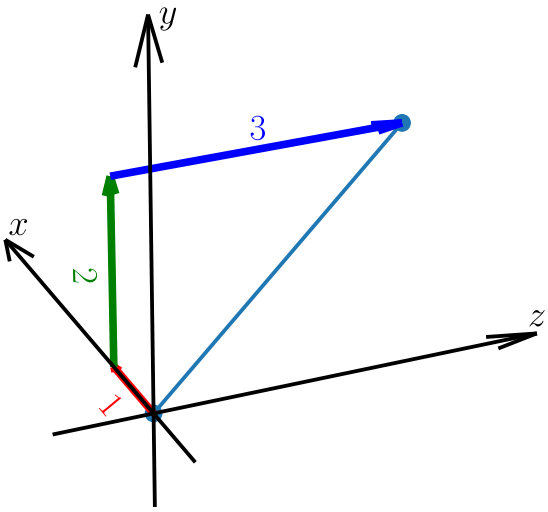


Figure 4: The vector $(1, 2, 3)$ broken down into its x, y, and z components

A **line** is a collection of points arranged in a straight path, extending infinitely in opposing directions.[3] In 2D, a line is most commonly defined as an equation conveying the relationship between the two coordinate variables x and y (1). Although this can be expanded to 3 dimensions (2), a more convenient and commonly used definition utilizes vectors (3). In this definition, every point P along the line is defined by adding a multiple t ($t \in \mathbb{R}$) of $\vec{V}$ to a point of origin O. Intuitively, it can be thought of as starting from an origin point O and traveling t amount in the direction of the vector component $\vec{V}$.

$$y = mx + b \tag{1}$$

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c} \tag{2}$$

$$P = O + t\vec{V} \tag{3}$$

Now that we have established a system of rules for representing space numerically, we can begin to lay the groundwork for projecting a 3D space onto a 2D one.

[3]Encyclopedia Britannica, *line*.

2.2 Viewports and Rays

The process of projecting a 3D view onto a 2D plane can be inspired by how vision and perspective works in the real world. Vision in the real world is essentially a 2D image of a 3D landscape.

Imagine you are looking onto a landscape and would like to save this view onto a canvas. An artist could simply paint the view, however you lack the skills required. Cleverly, you set up a net the same size as the canvas in a position that captures the view. Looking through a square hole of the net, you average out the color that you can see through that hole and paint the location corresponding to that hole on the canvas with that color. Repeating this step for every single hole results in an accurate sketch of the landscape.[4]

This metaphor is useful in that it can be applied to computers. Computers display images as a grid of square pixels, much like the square holes of the net used in the analogy.

A rectangle referred to as the **viewport** is placed in the world space, the equivalent to the square net. Then a camera is placed behind the viewport, the equivalent to your eyes (Figure-5).

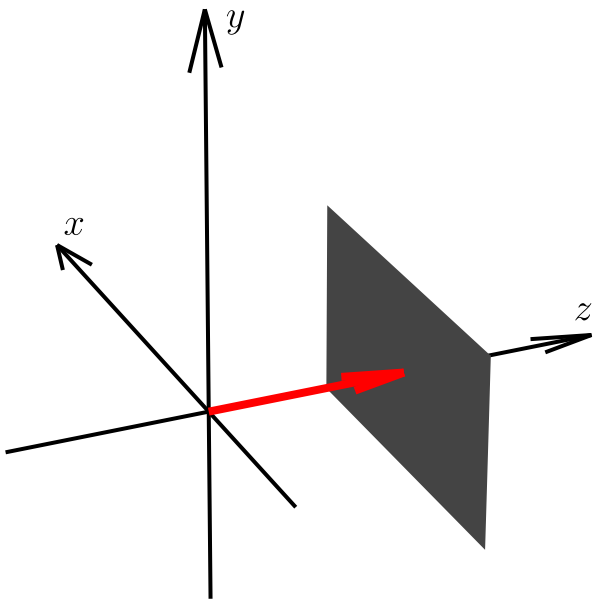


Figure 5: A viewport of a 16:9 screen with its center located at $(0, 0, 1)$

In the net analogy, the color that reaches your eyes are certain wavelengths of photons that travel from an origin light source, bounce off reflective objects, and ultimately end up in your retinas. Although we could simulate photons

[4]Gambetta, *Computer Graphics From Scratch*.

traveling in all direction from a light source, that would be highly redundant as most of them would not end up in your eyes. Instead, it is more efficient to trace in reverse: light **ray**s[5] traveling outwards from your eyes and into any objects.

In practice, this means that we compute shooting a light ray from the camera to a pixel's equivalent location on the viewport, and if it intersects with an object, color in that pixel on the screen with the color of the object. We repeat this for every pixel until every pixel on the screen is correctly colored in. This technique is known as **ray tracing**.

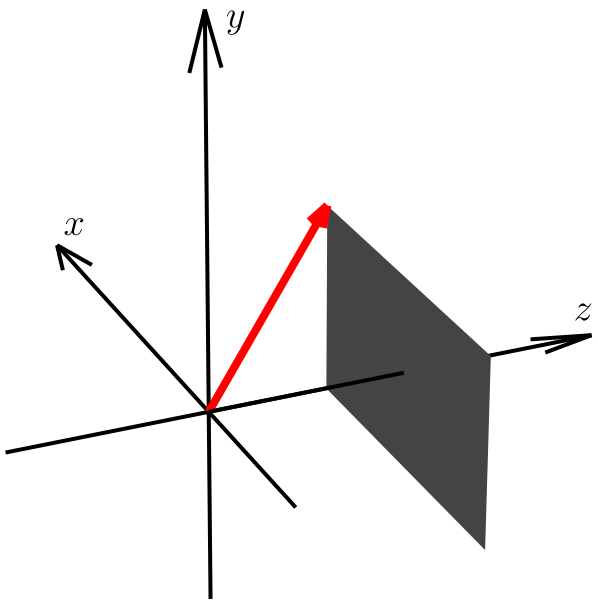


Figure 6: A vector from camera to pixel $(0, 0)$'s equivalent location on viewport

Now that a system for projecting a 3D space onto a 2D plane has been established, what remains is to implement said system.

[5]A ray is a collection of points arranged in a straight path extending infinitely towards a single direction. Whereas a line extends infinitely in two opposing directions from a point, rays only extend in a single direction.

2.3 A Ray Tracer Implementation

With a ray tracing system defined, an implementation can be programmed. The implementation in this paper will be coded in the C++ language, and the Raylib graphics library with the raylib-cpp wrapper will be used for drawing pixels onto a window.[6][7]

First, we must define a point, vector, and camera object (C++ class type).

Source Code 1: point.hpp

```
#ifndef POINT_H
#define POINT_H

struct Point {
    double x {};
    double y {};
    double z {};
};

static constexpr Point ORIGIN {0, 0, 0};

#endif
```

Source Code 2: vector.hpp

```
#ifndef VECTOR_H
#define VECTOR_H
#include "point.hpp"

struct Vector {
    double x {};
    double y {};
    double z {};
};

inline constexpr Vector operator+(const Vector& vector1, const
↪  Vector& vector2) {
    return {vector1.x + vector2.x,
            vector1.y + vector2.y,
            vector1.z + vector2.z};
}

inline constexpr Vector operator+(const Point& point, const
↪  Vector& vector) {
    return {point.x + vector.x,
            point.y + vector.y,
            point.z + vector.z};
}
```

[6] Santamaria, *Raylib*.
[7] Loach, *raylib-cpp*.

```
inline constexpr Vector& operator+=(Vector& vector1, const
↪  Vector& vector2) {
    vector1 = {
            vector1.x + vector2.x,
            vector1.y + vector2.y,
            vector1.z + vector2.z
        };
    return vector1;
}

inline constexpr Vector operator-(const Vector& vector1, const
↪  Vector& vector2) {
    return {vector1.x - vector2.x,
            vector1.y - vector2.y,
            vector1.z - vector2.z};
}

inline constexpr Vector operator-(const Vector& vector) {
    return {-vector.x,
            -vector.y,
            -vector.z};
}

inline constexpr Vector operator-(const Point& point1, const
↪  Point& point2) {
    return {point1.x - point2.x,
            point1.y - point2.y,
            point1.z - point2.z};
}

inline constexpr Vector operator*(double multiple, const Vector&
↪  vector) {
    return {multiple * vector.x,
            multiple * vector.y,
            multiple * vector.z};
}

inline constexpr double dotProduct(const Vector& vector1, const
↪  Vector& vector2) {
    return vector1.x * vector2.x +
    vector1.y * vector2.y +
    vector1.z * vector2.z;
}

inline constexpr Vector crossProduct(const Vector& vector1, const
↪  Vector& vector2) {
    double a1 = vector1.x;
    double a2 = vector1.y;
    double a3 = vector1.z;
```

```
    double b1 = vector2.x;
    double b2 = vector2.y;
    double b3 = vector2.z;

    return Vector{a2 * b3 - a3 * b2, a3 * b1 - a1 * b3, a1 * b2 -
    ↪  a2 * b1};
}

#endif
```

Source Code 3: camera.hpp

```
#ifndef CUSTOMCAMERA_H
#define CUSTOMCAMERA_H
#include <cmath>
#include "vector.hpp"

namespace Custom {
    class Camera {
        public:
            constexpr const Vector& cameraToViewportCenter()
            ↪  const {return m_cameraToViewportCenter;}

            constexpr const Point& pos() const {return m_pos;}

            constexpr double viewportWidth() const {return
            ↪  m_viewportWidth;}
            constexpr double viewportHeight() const {return
            ↪  m_viewportHeight;}

            constexpr int screenWidth() const {return
            ↪  m_screenWidth;}
            constexpr int screenHeight() const {return
            ↪  m_screenHeight;}

        private:
            Point m_pos {0, 0, 0};
            int m_screenWidth {960};
            int m_screenHeight {540};

            double m_viewportDistanceFromCamera {1};
            double m_viewportHeight {1};
            double m_viewportWidth
            ↪  {m_viewportHeight/m_screenHeight *
            ↪  m_screenWidth};
    };
}

#endif
```

Next, we must define a function to convert pixel coordinates to equivalent

in-world coordinates on the viewport. For now, we will assume the viewport center to be at $(0, 0, 1)$ and the camera to be at the origin $(0, 0, 0)$ for the sake of simplicity. Pixels on the screen are usually represented with a 2D cartesian coordinate system with an inverted Y-Axis (Figure-7).



Figure 7: Coordinate ranges for pixels on a $1920 \cdot 1080$ screen

For each pixel on the screen, its horizontal (x) distance from the center and vertical (y) distance from the center can be calculated.

```
double xDistanceFromCenter
↪  {camera.viewportWidth()/camera.screenWidth() * -(pixelX -
↪  camera.screenWidth()/2)};
double yDistanceFromCenter
↪  {camera.viewportHeight()/camera.screenHeight() * -(pixelY -
↪  camera.screenHeight()/2)};
```

Adding `xDistanceFromCenter` and `yDistanceFromCenter` to the x and y components of the vector going from camera to viewport center $(0, 0, 1)$ respectively results in a vector from the camera to the pixel's equivalent location on the viewport.

```
Vector{xDistanceFromCenter, yDistanceFromCenter, 1}
```

With this, a ray going starting from the camera position point (C) and going in the direction of the camera to pixel vector ($\vec{V}$) can be derived.

$$P = C + t\vec{V}$$

This is our light ray for the pixel (Figure-8).

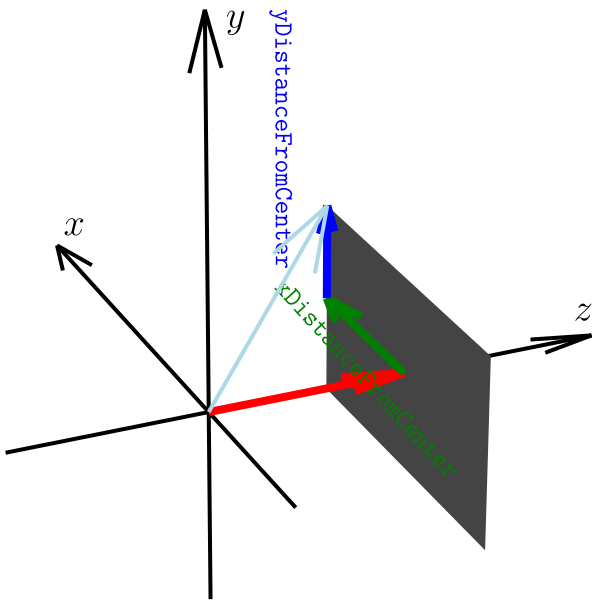


Figure 8

Operations on each pixel can be achieved through the use of a nested for loop.

```
for (int pixelX {0}; pixelX < 1920; ++pixelX) {
    for (int pixelY {0}; pixelY < 1080; ++pixelY) {
        <operation>
    }
}
```

References

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